

ZACHARY TIANYI TANG

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RESEARCH INTERESTS

Sample-efficient robot learning, human-in-the-loop adaptation, and precise/deformable-object manipulation.

EDUCATION

University of Toronto

Honours B.Sc., Computer Science; GPA: 3.77/4.00

Toronto, ON

Expected May 2028

RESEARCH & ENGINEERING EXPERIENCE

RLLAB, Yonsei University (Prof. Youngwoon Lee)

Research Intern

Seoul, South Korea

May 2026 - Aug 2026

- Conducted **self-directed research** on sample-efficient adaptation of pretrained **Vision-Language-Action (VLA)** policies using limited human feedback, studying when interventions most effectively correct robot behavior.
- Built and operated a **two-arm teleoperation setup** and helped develop its control framework for **DAGger-style intervention collection**, gathering real-robot operator demonstrations and human interventions.
- Implemented **real-time chunking (RTC)** for **diffusion policies**, using diffusion-space interpolation to blend overlapping action chunks for asynchronous robot control while maintaining policy performance.
- Implemented and evaluated **offline post-training (RL-Token-style)** using failure and intervention data, earning **top-8 finishes** in both tracks of the **RSS 2026 Robotics Foundation Models Challenge**.

Dynamic Optimization & RL Lab, University of Toronto (Prof. Chi-Guhn Lee)

Researcher

Toronto, ON

Sep 2025 - Present

- Co-authored **WireCraft (CoRL 2026 Spotlight; DOM-R3 @ IROS 2026 Oral)**, a **flexible-wire manipulation benchmark** showing that simulated plus real demonstrations outperformed real-only training on physical robots.
- Deployed, fine-tuned, and evaluated $\pi_{0.5}$ **VLA policies** with **OpenPI**, plus **ACT and multi-task DiT policies**, for UR5 wire manipulation; built **LeRobot-compatible datasets** and scalable training and inference pipelines.
- Built and benchmarked **Isaac Lab environments** for articulated and deformable wires, using **reinforcement learning with PPO**, reward shaping, clip-routing demonstrations, and task-success metrics for robot manipulation.
- Investigated rare **PhysX contact failures** across millions of RL steps with joint-state logging and **Three.js replay**; traced angular-velocity spikes to unstable contacts, informing mesh changes that made training substantially more stable.

Intelligent Robot Learning Lab, University of Alberta (Prof. Matthew Taylor)

Research Lead

Remote

Sep 2025 - Present

- Led an undergraduate research team studying how AI agents coordinate when each agent has incomplete information, building **JAX/JaxMARL training environments** and communication experiments.
- Built an **Isaac Lab simulation** with two **Franka robot arms** to study how robots share limited information while coordinating tasks, integrating high-level action controllers, local observations, and learned messaging.
- Received the **People's Choice Award** at the **2026 Canadian Undergraduate Conference on AI (CUCAI)**.

PROJECTS & LEADERSHIP

Heimer Academy | [React](#), [Vite](#), [FastAPI](#), [Supabase](#), [Amazon SageMaker](#), [Amazon Bedrock \(Devpost\)](#) ([GitHub](#))

- 1st Place — AWS × Riot Games Rift Rewind (\$10,000 USD + \$10,000 AWS credits)**
- Built a **League of Legends champion recommendation system** using gameplay patterns across **tens of thousands of player profiles** and similarity analysis over **600+ champion abilities**.

University of Toronto Machine Intelligence Student Team (UTMIST)

Director of Academics / Director of Research Collaborations

Toronto, ON

Jun 2025 - Sep 2026

- Established a research collaboration with the **University of Alberta** and led **15+ researchers** across two AI research teams.
- Benchmarked **TensorRT** video super-resolution inference latency, profiled GPU operations with **NVIDIA Nsight**, and built a **CUDA/OpenGL display path** that kept PyTorch tensors GPU-resident.

TECHNICAL SKILLS

Programming: Python, C, JavaScript, TypeScript

Robotics / ML: PyTorch, JAX, OpenPI, LeRobot, Isaac Sim, Isaac Lab, OpenCV

Systems / Tools: CUDA, TensorRT, SLURM, Docker, Git, Linux, Bash